Alejandro Gomez-Paz

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Education

University of Washington — Expected June 2028 — GPA: 3.8
Bachelor of Science in Applied Mathematics: Data Science

Coursework: Numerical Optimization (graduate), Probability I, Matrix Algebra, Scientific Computing, Data Programming, Statistical Reasoning, Formal Logic

Independent Study: Elements of Statistical Learning, Deep Learning, Statistics

Experience

Sensors, Energy, and Automation Laboratory (SEAL) — June 2025 – Present
Machine Learning Intern — Writeup

Built custom, production-ready RAG pipeline for RAD-AI: an embedded-AI product providing real-time natural language safety guidance from raw-sensor data to non-expert emergency responders

- Engineered a retrieval pipeline using a compute-optimal ranking algorithm, reducing hallucination 72%
- Built a vector database with semantic-aware chunking and optimal indexing across 30+ scraped sources, surfacing the correct source 29% more often
- Parametrized the architecture to tune retrieval against memory/compute hardware constraints, cutting tokens per query 34%
- Designed a 52-query benchmark suite derived from RAG research for objective validation

Highlighted Project

Explainable Fight Outcome Prediction Website for UFC — Live site | Writeup

A counter-factual classification model with confidence quantification and natural language reasoning served on a public website via automatic ETL pipeline

- Scraped, cleaned, and feature-engineered 3M+ datapoints, including a hyperparameter-tuned rating system increasing accuracy by 14% over baseline
- Trained a logistic regression model on 10,000+ fights, quantifying uncertainty with confidence intervals through a bootstrap ensemble
- Generated natural-language reasoning per prediction decomposed directly from model parameters
- Served from a public website backed by a self-updating ETL pipeline and cloud database with 100+ monthly visits

Skills

Skills: Optimization, RAG, LLMs, Supervised Learning (Regression and Classification), Model Selection and Cross-Validation, Unsupervised Learning (PCA, Clustering), Natural Language Processing, Feature Engineering, Statistical Analysis
Languages: Python, SQL, Tableau
Tools: pandas, NumPy, SciPy, scikit-learn, PyTorch, Matplotlib, HuggingFace, Jupyter Notebooks, Git/GitHub, LaTeX, Excel